# Chen Yang
World Models Researcher & Engineer
Principal Engineer at HUAWEI
*Last updated: November 2024*

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## Education

Sep 2021 – Jun 2025 **Ph.D. in Computer Science, Shanghai Jiao Tong University**
Ph.D. Advisor: ***Prof. Wei Shen***
Laboratory: ***Key Lab of Artificial Intelligence, Ministry of Education***

Sep 2019 – Jun 2021 **M.A. in Precision Instrument, Shanghai Jiao Tong University**
GPA: 3.76/4.0 | Top 10%

Sep 2015 – Jun 2019 **B.A. in Precision Instrument, Shanghai Jiao Tong University**
GPA: 3.91/4.3 | Top 30%

## Selected Research Projects

2024 **LiftImage3D: Lifting Any Single Image to 3D Gaussians with Video Generation Priors**
⋆ LiftImage3D is a framework that effectively releases the generative priors of latent video diffusion models (LVDMs) while ensuring 3D consistency; can lift arbitrary 2D images into high-quality 3D scenes composed of 3D Gaussians.
⋆ Project lead by ***Dr. Jiemin Fang*** *and* ***Dr. Qi Tian***.

2023 – 2024 **GaussianObject: High-Quality 3D Object Reconstruction from Four Views with Gaussian Splatting**
⋆ Aiming to reconstruct finely detailed objects from very sparse inputs (as few as 4 images). Leveraging 3DGS as scene representation and refining a pre-trained diffusion model for strong priors.
⋆ Project lead by ***Prof. Wei Shen*** and ***Dr. Jiemin Fang***.
⋆ Accepted by ***ACM Transactions on Graphics (TOG), SIGGRAPH Asia 2024***.

**EndoGSLAM: Real-Time Dense Reconstruction and Tracking in Endoscopic Surgeries using Gaussian Splatting**
⋆ Developing a novel SLAM approach for endoscopic surgeries that achieves over 100 fps rendering speed during online camera tracking and tissue reconstruction using streamlined 3D Gaussians.
⋆ Project lead by ***Prof. Wei Shen***.
⋆ Accepted by ***MICCAI 2024***.

2023 **Segment Anything in 3D with NeRFs**
⋆ Leveraging SAM (Segment Anything) to segment NeRFs, provide a generic method to lift 2D foundation models to the 3D space.
⋆ Project lead by ***Dr. Jiemin Fang*** *and* ***Prof. Wei Shen***.
⋆ Accepted by ***NeurIPS 2023***.

## Selected Research Projects (continued)

2022 – 2023 **Efficient Deformable Tissue Reconstruction via Orthogonal Neural Plane**

⋆ Accelerated the optimization and inference on reconstructing deformable tissues with NeRFs, improving efficiency and quality across non-rigid deformations.
⋆ Project lead by ***Prof. Wei Shen***.
⋆ Accepted by ***MICCAI 2023, Young Scientist Award*** and ***IEEE Transactions on Medical Imaging (TMI)***. Inspired **20+ publications** on dynamic tissue reconstruction.

2021 – 2022 **NeRFVS: Neural Radiance Fields for Free View Synthesis via Geometry Scaffolds**

⋆ Designed a novel approach enabling neural radiance fields to perform free view synthesis at room scale and perform superior extrapolation in room scale.
⋆ Project lead by ***Dr. Weichao Qiu*** and ***Prof. Wei Shen***.
⋆ Accepted by ***CVPR 2023***.

## Internship Experience

2023 – 2024 **3D Vision Intern, Huawei Cloud**, mentored by ***Dr. Jiemin Fang*** and ***Dr. Qi Tian***

⋆ Designed and implemented GaussianObject project, which enables high-quality 3D object reconstruction from very sparse inputs (as few as 4 images);
⋆ Paper accepted by SIGGRAPH Asia 2024 (TOG), work is open-sourced and widely recognized (***1,200+ GitHub Stars***).

2021 – 2022 **Machine Vision Intern, Huawei Noah's Ark Lab**, mentored by ***Dr. Weichao Qiu***

⋆ Designed and implemented NeRFVS project, significantly improving the extrapolation capability of neural radiance fields;
⋆ Proposed geometry scaffolds method, substantially enhancing scene reconstruction quality and extrapolation performance, paper accepted by CVPR 2023.

## Awards and Achievements

2023 **MICCAI Young Scientist Award**, Awarded top 5 among 2250 submissions.

**Intel Scholarship**, Awarded top 5 among over 100 competitors.

2022 **Second Prize of National Post-Graduate Mathematical Contest in Modeling**, Awarded to top 14.5% of contestants.

2021 **National Scholarship**, Awarded to top 3% of students at Shanghai Jiao Tong University.

**First Prize of Huawei Chinese University ICT Competition**, Awarded top 1 among 88 teams.

2019 – 2021 **First-class Academic Scholarship**, Awarded to top 30% of students at Shanghai Jiao Tong University.

## Research Publications


1. **C. Yang**[*], S. Li[*], J. Fang, et al., "Gaussianobject: Just taking four images to get a high-quality 3d object with gaussian splatting," ***ACM TOG***, 2024.
2. **C. Yang**, K. Wang, Y. Wang, X. Yang, and W. Shen, "Neural lerplane representations for fast 4d reconstruction of deformable tissues," in ***MICCAI***, ***Young Scientist Award***, Springer Nature Switzerland Cham, 2023, pp. 46–56.

3. **C. Yang**, P. Li, Z. Zhou, et al., "Nerfvs: Neural radiance fields for free view synthesis via geometry scaffolds," in ***CVPR***, 2023, pp. 16 549–16 558.
4. **C. Yang**, K. Wang, Y. Wang, Q. Dou, X. Yang, and W. Shen, "Efficient deformable tissue reconstruction via orthogonal neural plane," ***IEEE TMI***, 2024.
5. **C. Yang***, K. Wang*, Y. Wang, et al., "Endogslam: Real-time dense reconstruction and tracking in endoscopic surgeries using gaussian splatting," in ***MICCAI***, Springer, 2024, pp. 219–229.
6. **C. Yang**, S.-Y. Yao, Z.-W. Zhou, B. Ji, G.-T. Zhai, and W. Shen, "Poxture: Human posture imitation using neural texture," ***IEEE TCSVT***, vol. 32, no. 12, pp. 8537–8549, 2022.
7. **C. Yang***, H. Zhao*, H. Wang, and W. Shen, "Chase: 3d-consistent human avatars with sparse inputs via gaussian splatting and contrastive learning," arXiv preprint arXiv:2408.09663, 2024.
8. K. Wang*, **C. Yang***, K. Zhao, X. Yang, and W. Shen, "Realistic surgical simulation from monocular videos," arXiv preprint, 2024.
9. Y. Chen*, **C. Yang***, J. Fang, et al., "Liftimage3d: Lifting any single image to 3d gaussians with video generation priors," arXiv preprint, 2024.
10. H. Zhao*, **C. Yang***, H. Wang, X. Zhao, and W. Shen, "Sg-gs: Photo-realistic animatable human avatars with semantically-guided gaussian splatting," arXiv preprint arXiv:2408.09665, 2024.
11. B. Ji, **C. Yang**, Y. Shunyu, and Y. Pan, "Hpof: 3d human pose recovery from monocular video with optical flow," in ***ICMR***, 2021, pp. 144–154.
12. P. Li, S. Wang, **C. Yang**, B. Liu, W. Qiu, and H. Wang, "Nerf-ms: Neural radiance fields with multi-sequence," in ***ICCV***, 2023, pp. 18 591–18 600.
13. J. Cen, J. Fang, **C. Yang**, et al., "Segment any 3d gaussians," in ***AAAI***, 2024.
14. R. Liang, J. Zhang, H. Li, **C. Yang**, Y. Guan, and N. Vijaykumar, "Spidr: Sdf-based neural point fields for illumination and deformation," ***CVPRW***, 2023.
15. Y. Yan, Z. Zhou, Z. Wang, **C. Yang**, J. Gao, and X. Yang, "Dialoguenerf: Towards realistic avatar face-to-face conversation video generation," ***Visual Intelligence***, vol. 2, no. 1, p. 24, 2024.
16. J. Cen, Z. Zhou, J. Fang, **C. Yang**, et al., "Segment anything in 3d with nerfs," ***NeurIPS***, vol. 36, pp. 25 971–25 990, 2023.
17. J. Lu, T. Yi, J. Fang, **C. Yang**, et al., "Snap-snap: Taking two images to reconstruct 3d human gaussians in milliseconds," arXiv preprint, 2024.
18. Z. Zhou, R. Zhong, **C. Yang**, Y. Wang, X. Yang, and W. Shen, "A k-variate time series is worth k words: Evolution of the vanilla transformer architecture for long-term multivariate time series forecasting," arXiv preprint arXiv:2212.02789, 2022.

## Skills

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| Programming Languages | Python, C, C++, Matlab |
| Software & Tools | PyTorch, OpenCV, OpenGL, LaTeX, Jax, COMSOL |

## Academic Services

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| Conference Reviewer | CVPR '23, '24; ICCV '23; NeurIPS '23, '24; ECCV '24; AAAI '25; MICCAI '23, '24 |

## Academic Services (continued)

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| Journal Reviewer | TOG; TMI; TCSVT; TOMM |
| Teaching Assistant | Spr. 2019: **MI 321**: Course Design of Instrument Bus and Virtual Env.<br>⋆ Guided students in bus programming and virtual instrument development projects.<br>Fa. 2020: **MI 318**: Measuring and Controlling Circuit<br>⋆ Facilitated lab sessions and assisted students with hands-on circuit experiences.<br>Spr. 2021: **EE 334**: Industrial Measurement and Control Tech. and Sys.<br>⋆ Guided students through industrial measurement processes and control technologies. |